How to Read a Paper

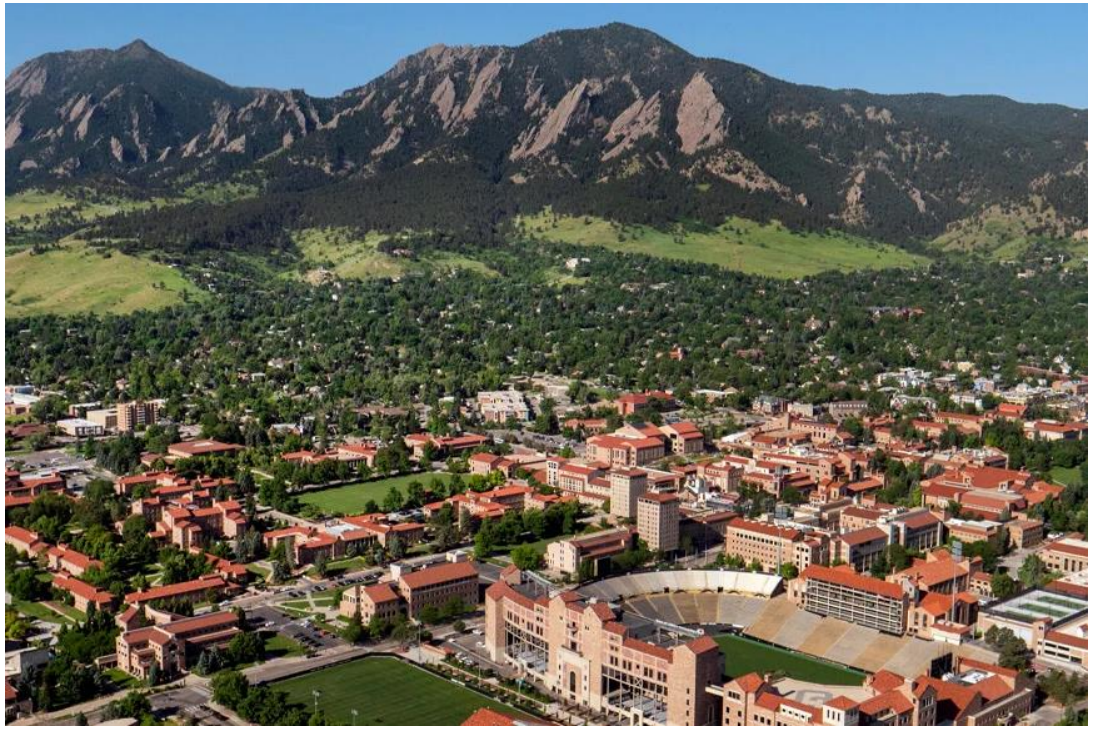
Course: Networking Principles in Practice – Cloud Networking
Module: Intro



University of Colorado **Boulder**

Sorry for the text heavy slides

(here’s a couple pics to make up for it)



Why are we reading academic papers?

- Modern cloud infrastructure required innovation to create
- Academic papers are how those innovations are communicated to the world
  - E.g., Software-defined networking came from a bunch of academics and publications across a number of years
- Cloud providers (esp. Google and Microsoft) have a strong commitment to the research communities
  - And publish a lot of their work
- Key network systems conferences: SIGCOMM, NSDI, CoNEXT
- Other venues for systems: SOSP, ASPLOS, OSDI, Usenix ATC, Usenix Security, IEEE Security and Privacy, (and many more)

Main common components of a paper

- Abstract – short 1-2 paragraph overview
- Intro – a summary of the entire paper. Why it's important, what's the alternative, what's the insight, what's the system, and what's the findings
- Motivation / Background / Related Work – provides the necessary context for understanding the paper and its novelty
- Architecture – an overview of the system
- Implementation – details about how the architecture was realized
- Evaluation – proves the claims, and alleviates any potential concerns. Micro benchmarks focus on key details, Macro benchmarks focus on the system working as a whole
- Discussion – Any extra notes that wasn't addressed, may be a concern and the authors want to suggest how they'd address it, or may be an extra use of it that's not proven out
- Conclusions and Future Work – summarize the paper again, and add in what might be some next steps

Some example approaches to reading a paper

There may be different reasons for reading a paper, so there's different approaches that people could take.

We'll focus on published work where the reader is looking to learn (as opposed to reviewing something to be critical and improve the paper)

- Intro + architecture figures
- Focus on background / related work
- Really deep, with particular focus on evaluation (what systems/datasets did they use, how did they prove their point)

Industry papers

- Some have less of a feel of introducing something new
- But, really useful to see what approaches they took and background they built on
- Really useful to see what worked or was challenging at scale
- Gives important context for challenges that still need to be solved



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